



Operational Signals & KPIs

Week 2 • Day 1 • Strategic Design & Metrics

TPM Academy



Day 1 Objectives

- Distinguish **operational signals**, **key performance indicators (KPIs)**, and a **North Star** — three different audiences, three different rhythms
- Spot vanity metrics by their two telltale signs: *easy to move*, *hard to attribute*
- Build a believable **causal chain** from operational signal → KPI → NS
- Identify the **instrumentation gaps** that block your strategy from being measurable today
- Produce a quad **Metrics Tier Sheet** for your Week-1 problem statement



Today's Shape

Clock	Block	Outcome
09:00 – 09:15	Opening	Reground in your Week-1 problem statement
09:15 – 10:30	Block 1 + Activity 1	"Metric or Vanity?"
10:45 – 12:00	Block 2 + Activity 2	Operational signal sprint
13:00 – 14:15	Block 3 + Activity 3	KPI laddering
14:30 – 16:00	Block 4 + Activity 4 + Readouts	Tier Sheet defense

Block 1 — Three Tiers, Three Audiences

Why we say "metric" too loosely



The Metrics Pyramid

North Star — "Are we winning?"

KPIs — "Is the business healthy?"

Operational signals — "What do we do today?"

The shape matters: few NSs (1), few KPIs (2–4), many operational signals (4–10). A team that has 10 KPIs has 0 KPIs.



Definitions That Hold Up Under Pressure

Tier	Definition	Rhythm	Audience
North Star	The single number that says "the customer is winning AND the business is winning"	Quarterly to annual	Whole company
KPI	Business outcome that the NS rolls up to or directly signals	Weekly to monthly	Leadership / board
Operational signal	Daily/hourly leading indicator a squad can act on	Real-time to daily	Squad / TPM

The TPM job: own the operational signals, defend the KPIs, contribute to the North Star.



The Vanity Metric Trap

A vanity metric is one that **goes up** and looks impressive, but whose rise **can't be attributed** to a deliberate decision that created customer value — easy to move, hard to attribute.

✗ Vanity

- Total signups (no churn signal)
- Page views
- Features shipped
- Average session length

✓ Actionable

- Activated accounts in week 1
- Time-to-first-value
- % of intended workflows completed
- Repeat-use within 7 days

Activity 1 — "Metric or Vanity?"

Quads • 45 minutes

Each quad receives the **Vanity-Metric Card Pack** (12 cards from FieldPulse-adjacent SaaS). Place each on a 2x2 of *Easy-to-game* vs *Lagging/Leading*.

- Pick 2 cards your quad would **bet on** for FieldPulse
- Pick 2 cards your quad would **never report up**
- Be ready with one sentence on each pick



10 min sort • 15 min discuss • 10 min readout prep

Block 2 — Operational Signals

The metrics a squad watches in standup



A Good Operational Signal

- **Latency:** moves within a day or less
- **Specificity:** changes when *one thing* in the product changes
- **Actionability:** if it drops 30%, the team has a meeting tomorrow
- **Causal:** plausibly rolls up to a KPI
- **Measurable today** — or honestly named as an *instrumentation gap*

Coach's note: "We don't measure that today" is not a failure — it's a finding. Surface it.

FieldPulse Worked Example

Problem statement (Week 1): "Dispatchers spend 45 minutes after-shift reconciling tickets, truck stock, and timecards."

Operational signal	Source	Action if it dips
Median reconcile-flow duration	App instrumentation	Inspect last release for regressions
% of dispatchers using mobile reconcile vs paper	Backend events	Re-investigate adoption blockers
Reconcile-flow abandonment rate	Funnel analytics	UX review of last drop-off step
Manual override entries per dispatcher per day	<i>Instrumentation gap</i>	Indicates trust loss in automation

Activity 2 — Operational Signal Sprint

Quads • 50 minutes

For your quad's Week-1 problem statement: brainstorm 12 candidate signals (4 per person), then cull to **6 finalists**.

- For each: what it measures, where the data comes from, how fast it moves, what action you'd take
- Mark **at least 2 in red** as instrumentation gaps
- Add the 6 to your quad's **Metrics Tier Sheet** (bottom row)

 10 min solo • 15 min cull • 15 min annotate

Block 3 — KPIs

Where the squad's work shows up in business outcomes

The Four-Question KPI Quality Bar

A well-formed KPI passes all four:

1. Spoken in **business terms** (revenue, retention, time-to-value, Net Promoter Score / NPS) — not feature terms
2. A senior leader cares about it **without translation**
3. You can imagine a **single chart** that shows it weekly
4. You can name what's "good" **without inventing a number** — direction is fine, the number is set later



Laddering: Signals Up to KPIs

Step

- Take each operational signal
- Ask: "If this moves 30% for a quarter, which business outcome shifts?"
- Cluster signals that ladder to the same outcome
- Each cluster → candidate KPI

Worked: FieldPulse

- Reconcile-duration + abandonment + manual-overrides → **"Dispatcher hours returned per week"**
- Mobile-reconcile adoption → **"% of accounts on the new workflow"**
- Both ladder to **net revenue retention (NRR)** long-term

The Believable Causal Chain

Your Tier Sheet is only as strong as its narrative: **operational signal** → **KPI** → **North Star**. If you can't tell that story in three sentences, the chain is broken.

FieldPulse three-sentence chain:

"Median reconcile-flow duration drops → dispatchers spend less time on after-shift admin (operational). Hours-returned-per-week climbs → account-level retention conversations get easier (KPI). Net retention rate climbs → FieldPulse is winning (North Star)."

Activity 3 — KPI Laddering

Quads • 50 minutes

Cluster your 6 operational signals into **2–3 KPI clusters**. Stress-test each KPI against the four-question quality bar.

- Write the three-sentence causal chain for each cluster
- Name one signal that *doesn't* ladder to anything — and decide whether to keep or drop it
- Add KPIs to your **Metrics Tier Sheet** (middle row)



15 min cluster • 15 min stress-test • 10 min chain narration

Block 4 — Tier Sheet Defense

Before the squad ships, the metrics get challenged



Three Standard Challenges

Challenge	Question	What it surfaces
Causal chain	"Show me how a change in op-signal #3 plausibly moves your KPI"	Hand-waved laddering
Gameability	"Pick the easiest of these to game. How would a sales-incentivized team game it?"	Metrics that reward bad behavior
Instrumentation	"Which two of these can you actually measure today, in production?"	Strategy that depends on data we don't have

Watch for: "We'll instrument it" — a real answer also says *by when* and *at what cost*.

★ The North Star Placeholder

You're not going to nail your North Star today. Tomorrow is its full treatment. Today: write a placeholder.

```
We are winning when [customer outcome] increases for  
[user segment], measured by [single number].
```

FieldPulse example placeholder:

"We are winning when **after-shift reconcile time** drops for **HVAC dispatchers in mid-sized shops**, measured by **median minutes from clock-out to ticket-closeout**."

Activity 4 — Tier Sheet Defense

Quads • 55 minutes • Readouts

Pair-defense round: each quad pairs with another. Defender presents the Tier Sheet (4 min). Challenger uses the **Three Standard Challenges**.

- 15 min: draft your **North Star placeholder** using the template
- 20 min: pair-defense (4 min present, 4 min challenge per direction)
- 10 min: capture the strongest challenge as a "known weakness" line

 15 min NS • 20 min defense • 10 min refine



Day 1 Wrap

What we built

- Vocabulary: NS / KPI / Operational signal
- Vanity-metric detector
- Three-sentence causal chain
- Quad Metrics Tier Sheet

What opens tomorrow

- **The North Star metric** — done properly, with three pitfalls
- Your placeholder NS will be challenged. Bring an alternative.

Bring to Day 2: Your Tier Sheet, your NS placeholder, and one alternative NS your quad debated and rejected.

Questions & Reflection

Which metric on your Tier Sheet would your customer most want to see drop?